

Anatomy and Physiology of Female Reproductive System

The Female Pelvis

Learning Objective

At end of this session student be able to

- ✓ Define Female bony pelvis
- ✓ List function of pelvis
- ✓ Describe Part of female pelvis
- ✓ Describe measurement of female pelvis
- ✓ List type of female pelvis

Parts of the Bony Pelvis

- Bony pelvis divided by the into:
 - False (greater)
 - True (lesser) pelvis
- Classification of birth canal:
 - Pelvic inlet
 - Mid-pelvis
 - pelvic outlet

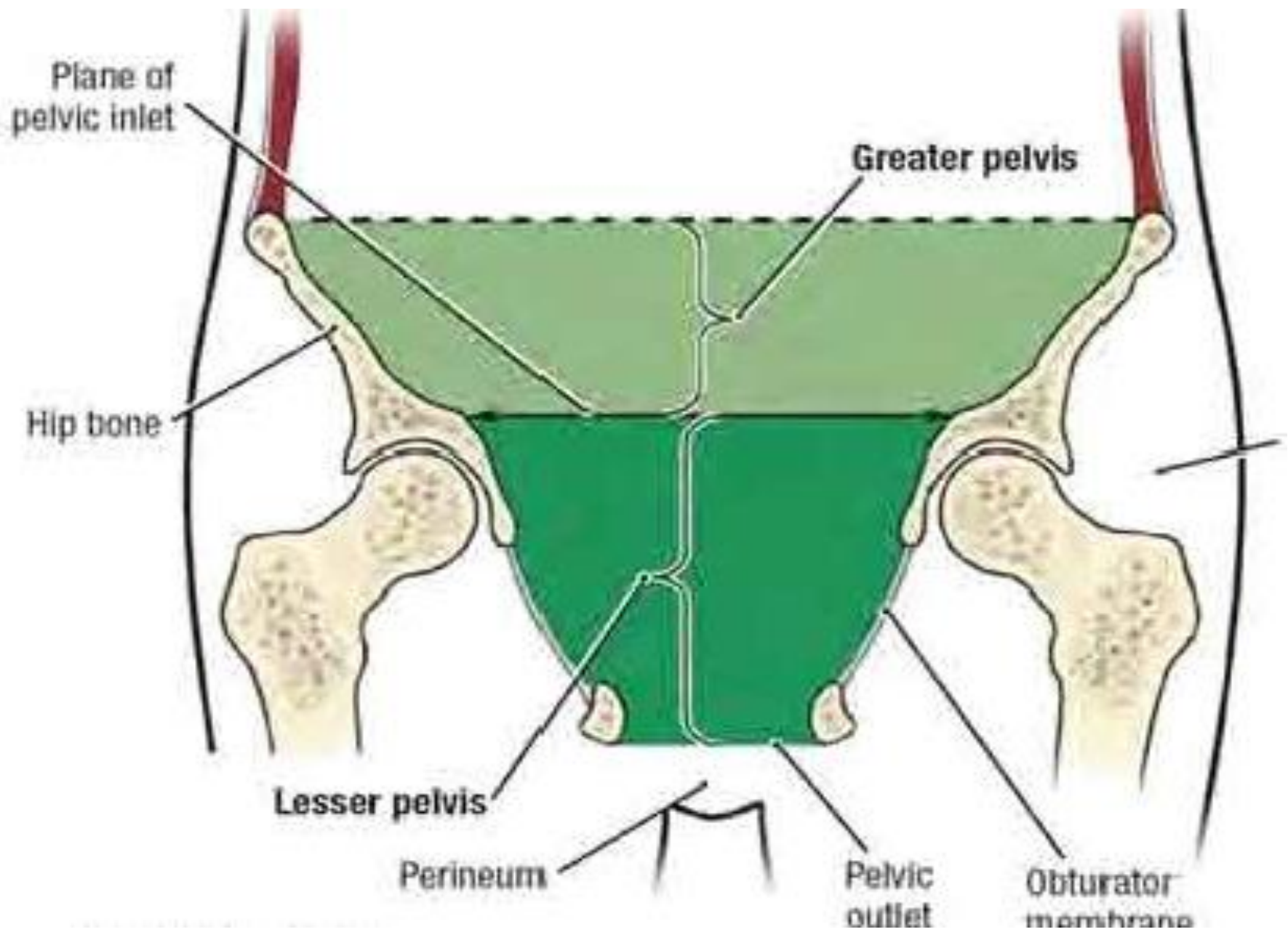
Division of pelvis

1. False Pelvis

- Lies above the true pelvis.
- It is formed by the upper flared-out portions of the iliac bones
- Protects the abdominal organs
- Has no significance in obstetrics

Cont...

- Its boundaries are:
 - **Anteriorly:** Anterior abdominal wall
 - **Laterally:** the iliac fossae
- **Posterior:**
 - ✓ lumbar vertebrae



2. True Pelvis:

- It is the bony canal through which the fetus passes during birth.

It has Three parts:

1. Brim or inlet
2. Mid cavity and
3. Outlet

Cont...

- ✓ **Pelvis:** is the wide curved set of bones at the bottom of the body that the *leg and spine are connected*
- ✓ A part of the trunk infero-posterior to the abdomen
- ✓ Area of transition between the trunk and the lower limb
- ✓ Space or compartment surrounded by the pelvic girdle (bony pelvis)

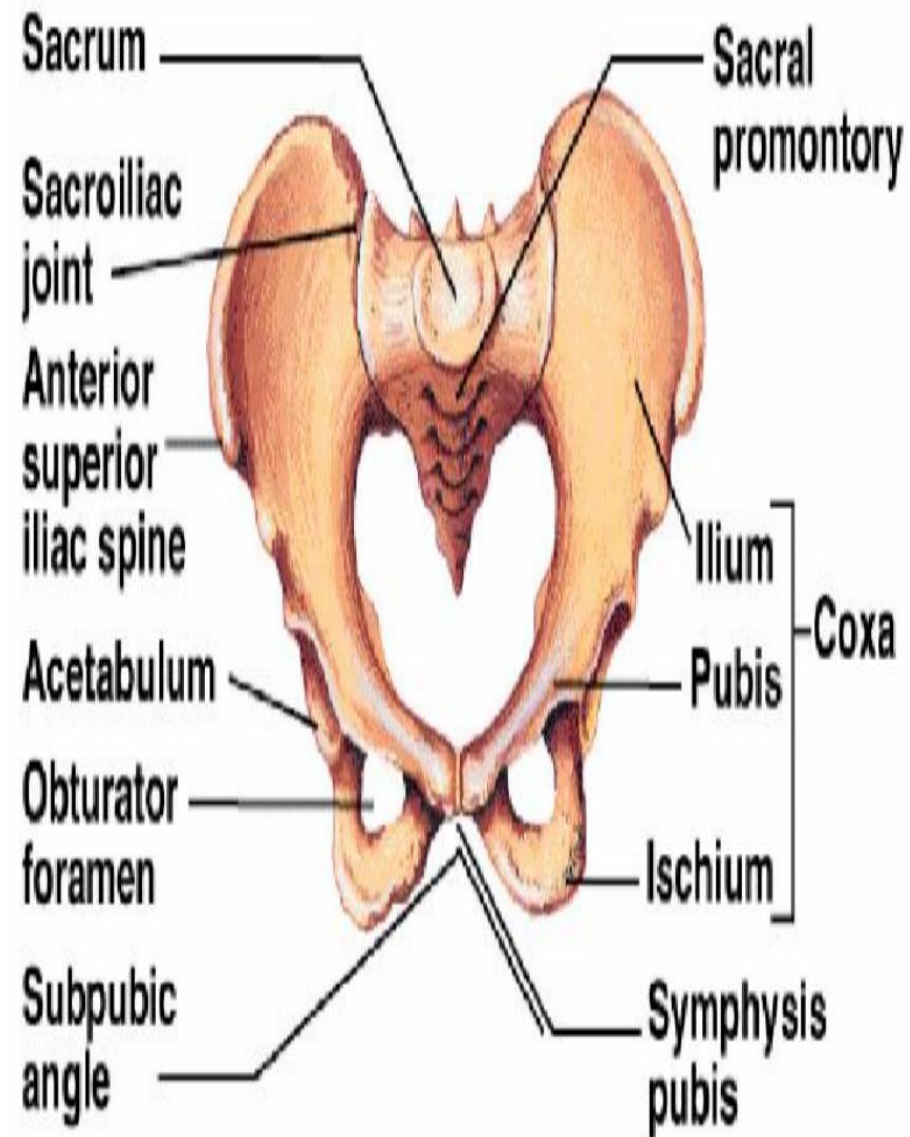
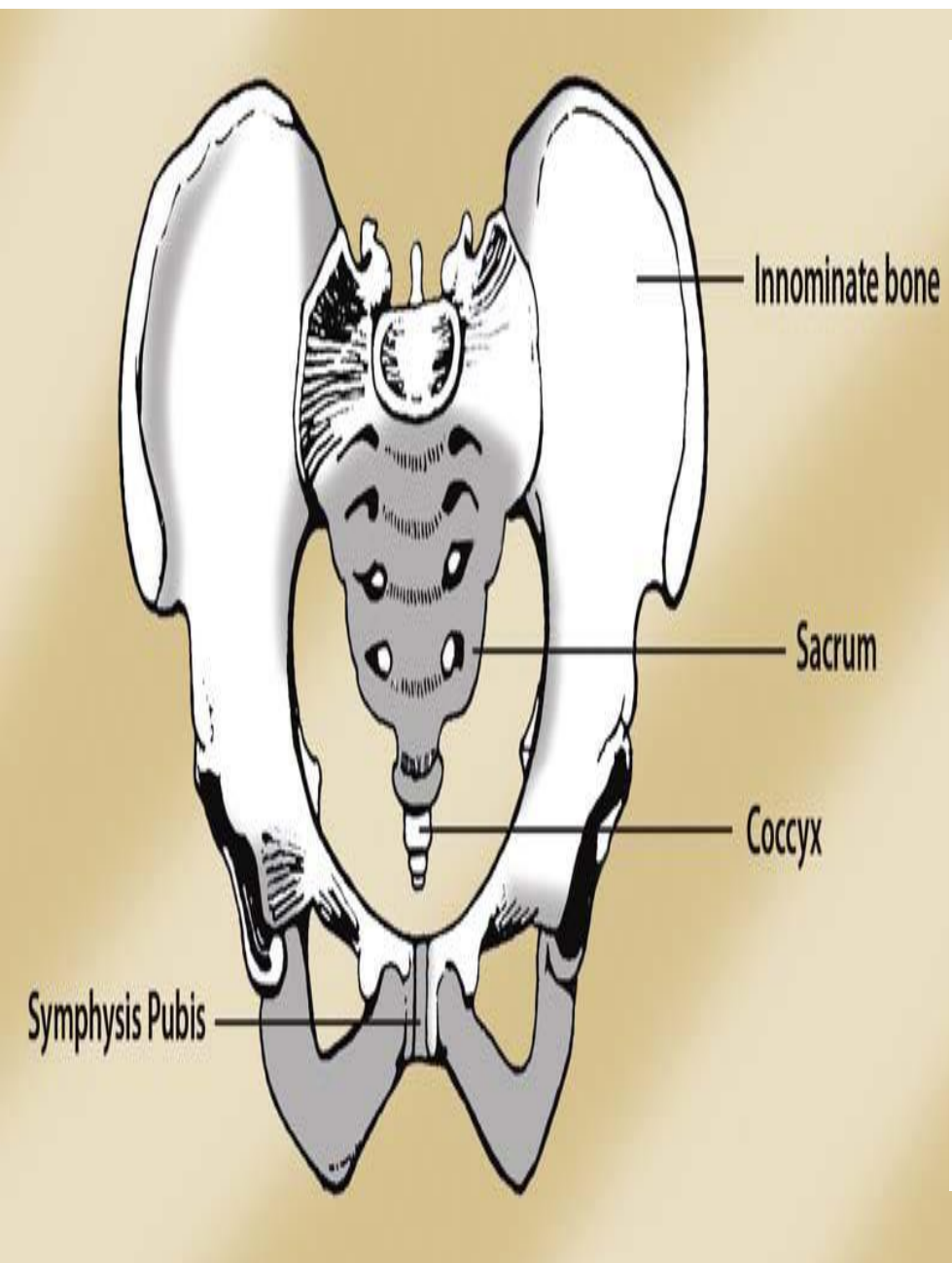
Functions

- ✓ Protection of the pelvic structures
- ✓ Accommodation of the growing fetus during pregnancy
- ✓ Anchorage of the pelvic support structures.

Cont...

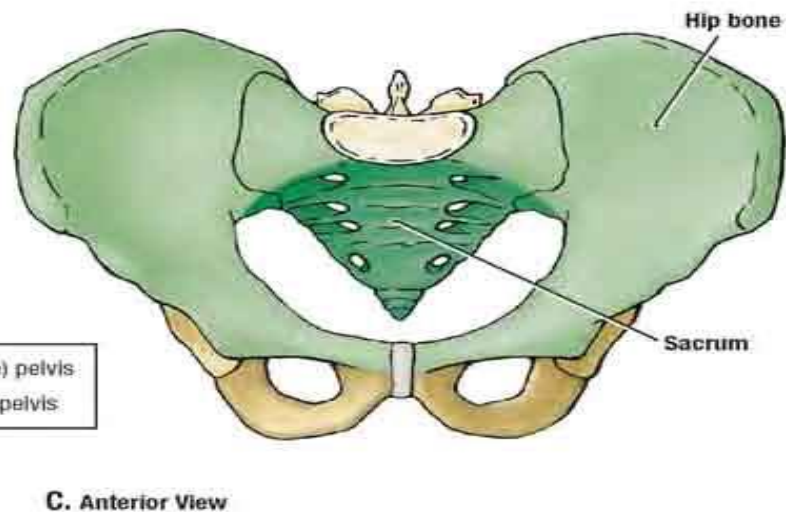
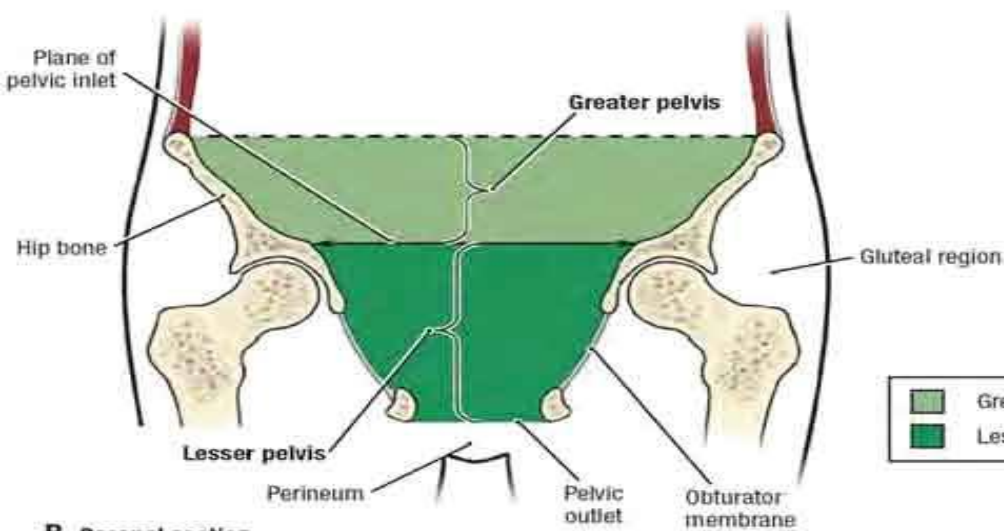
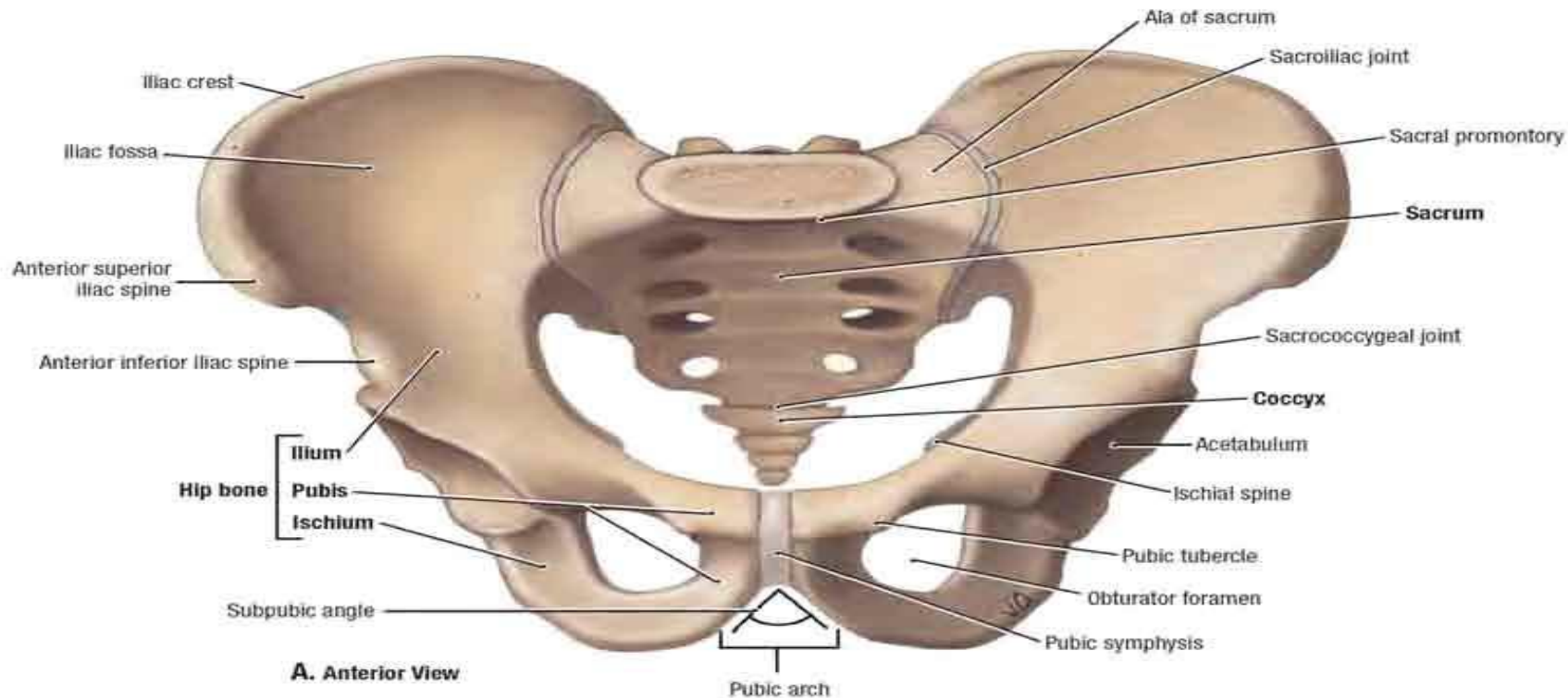
The pelvis is made of three bones:

- Two innominate or hip bones,
- One sacrum
- One coccyx



1. Innominate Bones:

- Are placed laterally and anteriorly.
- Each is formed by fusion of
 - Ilium
 - Ischium
 - Pubic bones around the acetabulum



Ilium

➤ Filared - Out part of innominate bone

It consists of:

- **Iliac crest:** ridge of ilium
- **Iliac fossa:** The large shallow depression that forms much of the inner surface of the ilium above the pelvic brim

Ischium

- Thickest posterior- inferior part of the hip bones.

It has two parts

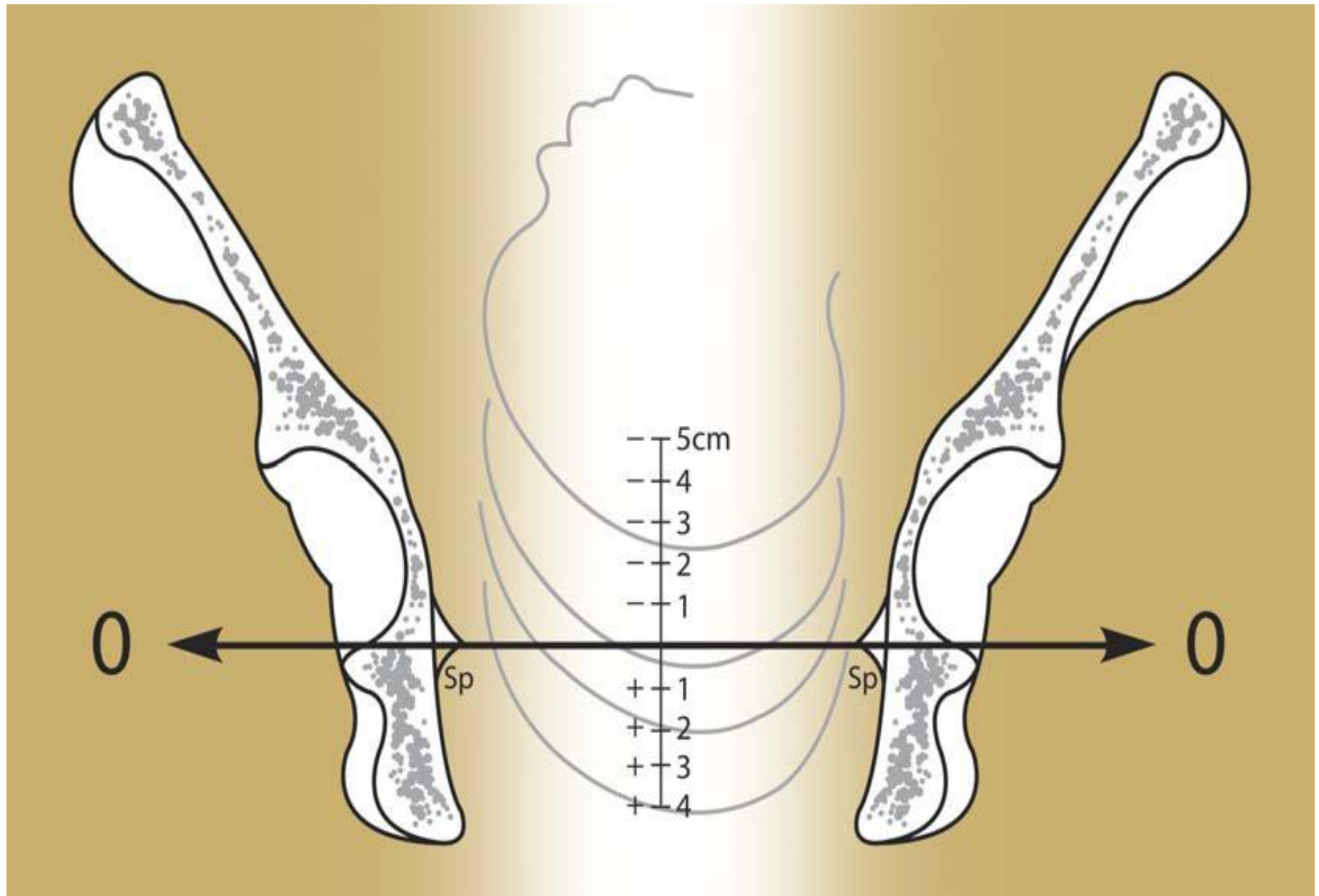
1. Ischial tuberosity:-

- Is the lower part of the innominate bone
- It is a large prominent where the body rests when in sitting position

Cont....

2. Ischial spine:-

- A projection superior to the tuberosity
- It is useful land mark in making vaginal examination i.e. the **station**.
- From this the outlet of true pelvis is started



Pubis

- Forms anterior part of the hip bone
- It meets the pubis of the opposite side in the cartilaginous joint called ***Symphysis pubis***.

It has two rami:-

- + Superior ramus
- + Inferior ramus which forms the pubic arc

2. Sacrum bone

- Triangular /wedge-shaped bone
- It consists of 5 fused vertebrae

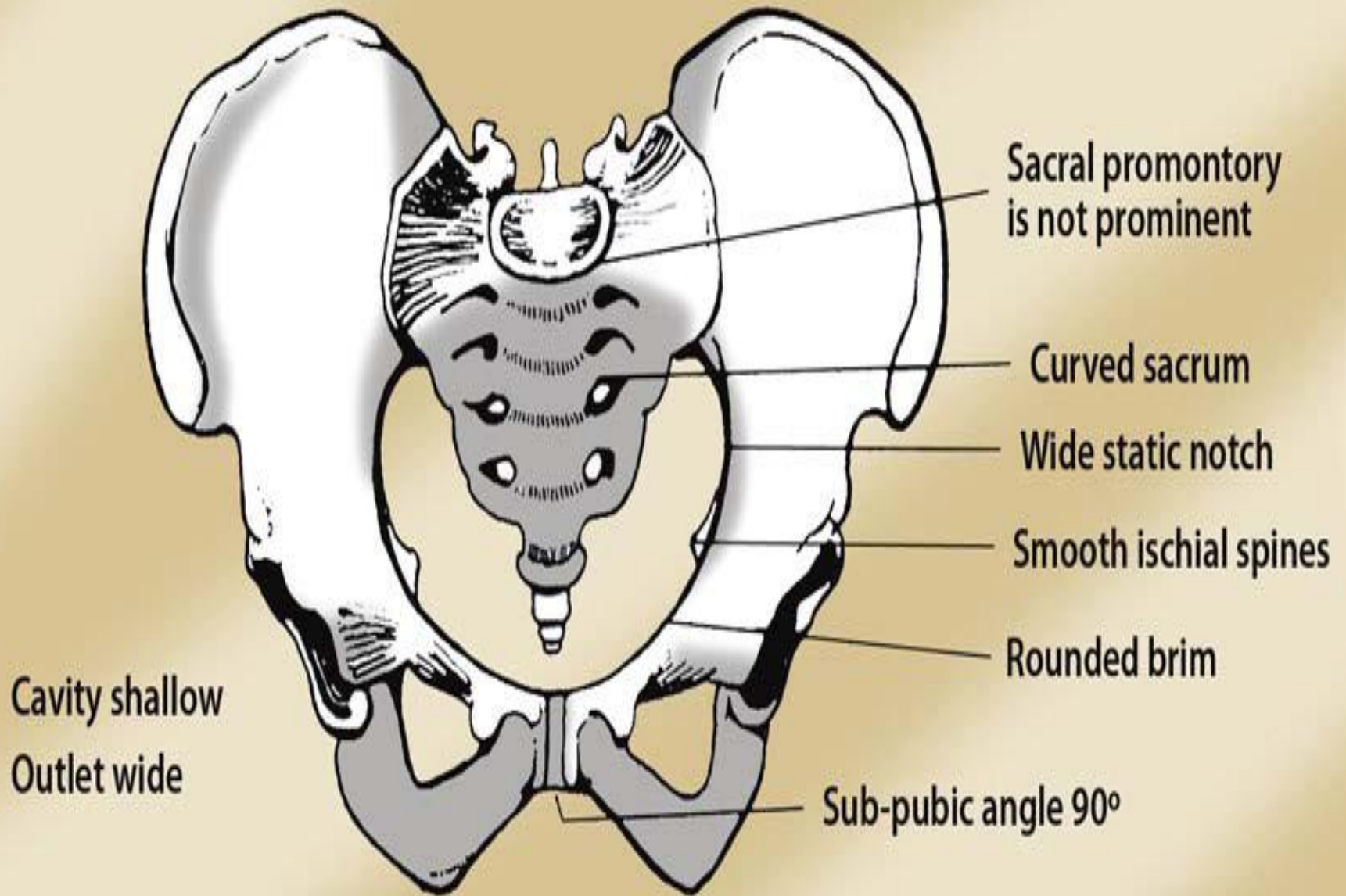
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Parts:-

-  **Sacral promontory:** - it is the upper margin and an important landmark of the pelvis.
-  **Hollow of the sacrum**-the smooth concave anterior surface.
-  **Sacral ala or wing** –the area on either side of the promontory.

3. Coccyx

- It is a small triangular shaped bone consisting of **4 vertebrate bones fused together.**
- During labor the coccyx moves back ward to enlarge outlet



Crest of ilium

Anterior superior iliac spine

Posterior superior iliac spine

Anterior inferior iliac spine

Posterior inferior iliac spine

Greater sciatic notch

Acetabulum

Ischial spine

Superior ramus of pubic bone

Lesser sciatic notch

Symphysis pubis

Ischial tuberosity

Inferior ramus of pubic bone

Obturator foramen

Inferior ramus of ischium

Pelvic joints

- There are four joints
 - 1. Symphysis pubis (1)** ;-between two pubic bones
 - 2. Sacroiliac joint (2)** ;-between sacrum & ileum(strongest joint)
 - 3. Sacrococcygeal joint (1)** :-between sacrum & coccyx

N.B During pregnancy the endocrine activity causes the ligaments to soften which allows the joints to give way & provide more room for the fetal head to come out

The pelvic brim(inlet)

- The pelvic brim separates the 'false' pelvis from the 'true' pelvis.
- It is a round shape, with the exception of the sacral.

Its bounded by:

- Anterior border: the pubic bones
- Lateral borders: the iliac bones
- Posterior border: the promontory and wings of the sacrum

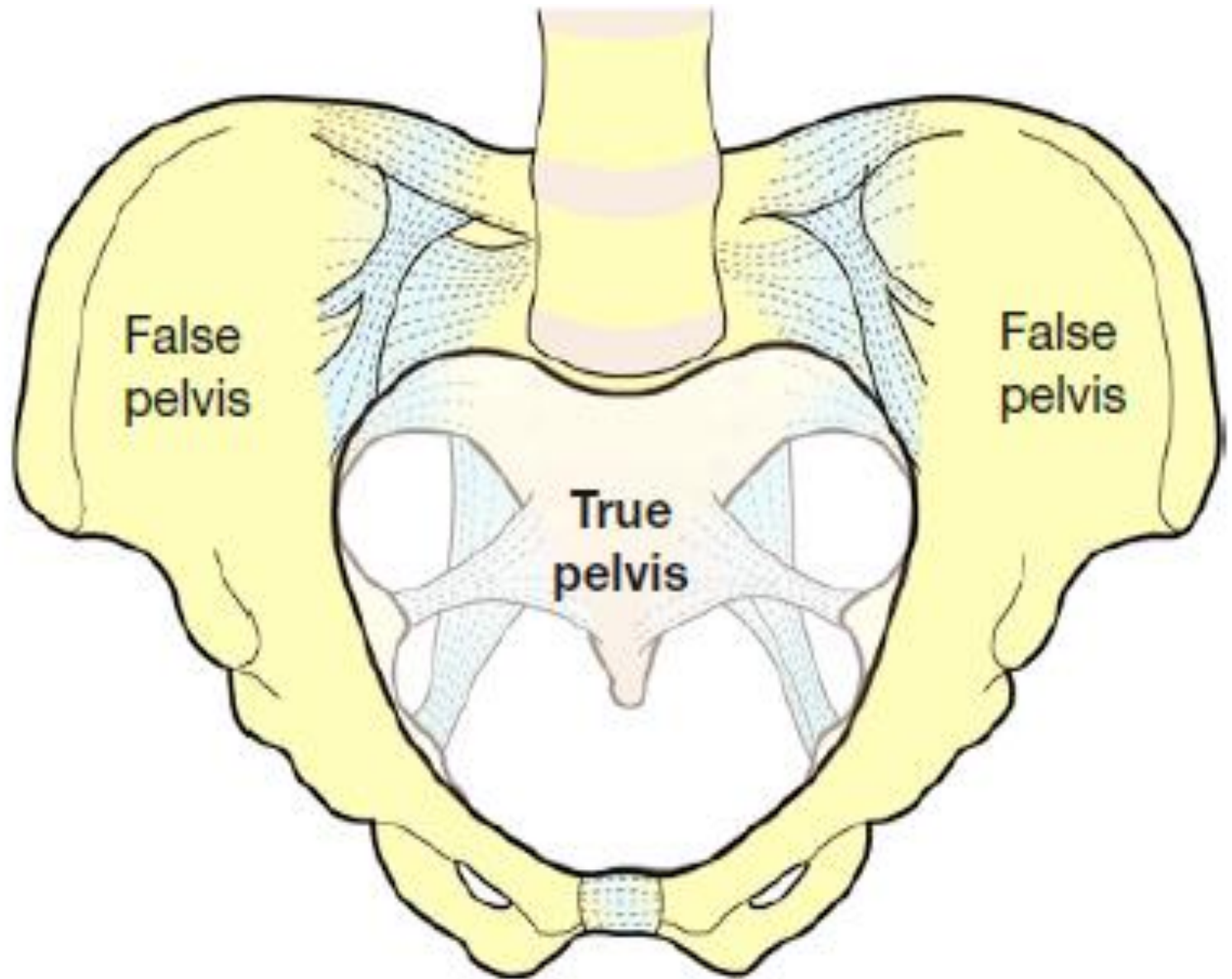
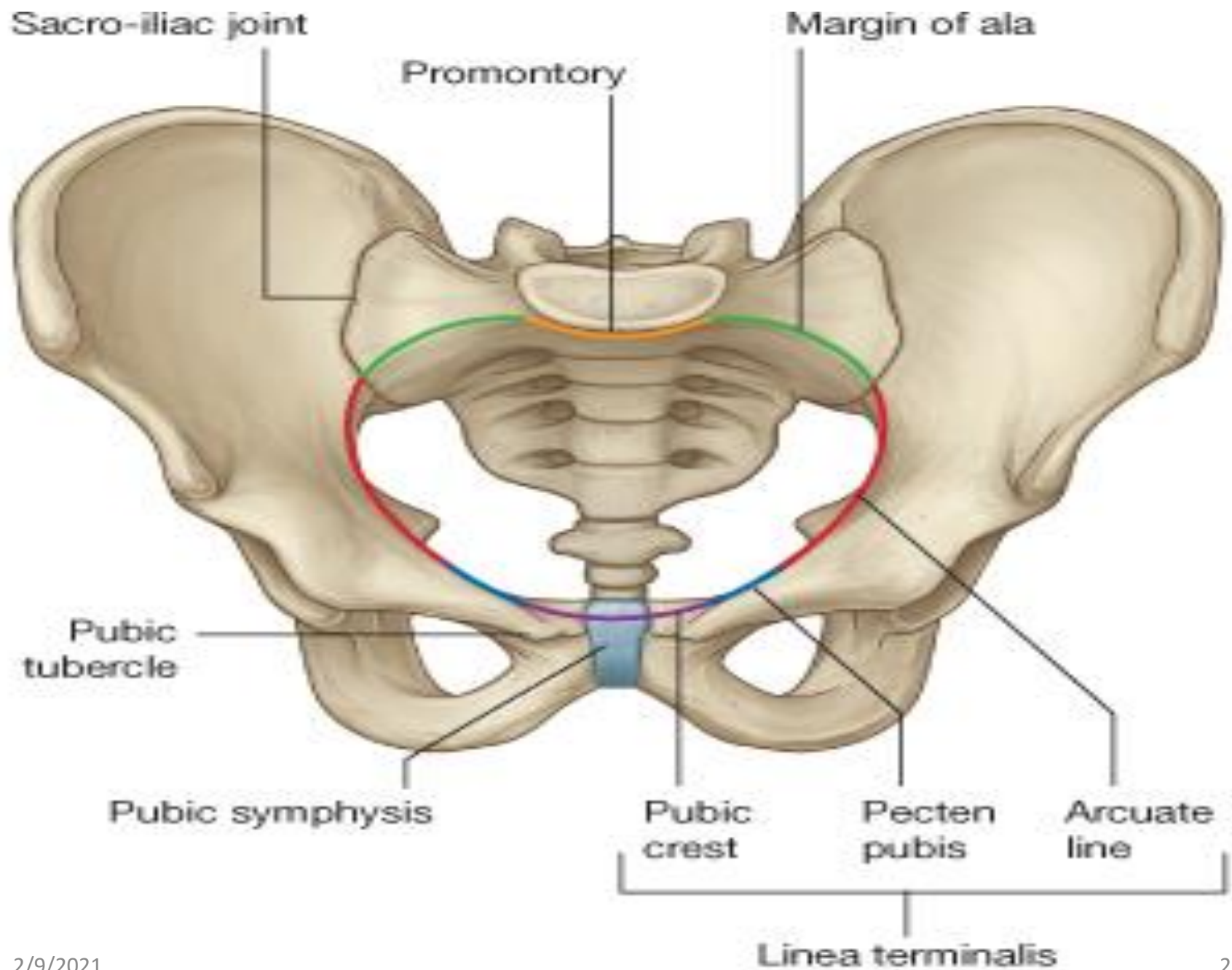


Fig. 1.1 The 'true' and 'false' pelvis.

Landmarks and fixed parts of the pelvic brim:

- Sacral promontory
- Sacral ala
- Sacroiliac joint
- Iliopectineal line
- Superior ramus of the pubic bone
- Upper inner border of the body of the pubic bone
- Upper inner border of the symphysis pubis



Diameter of Pelvic Brim

A. Anterior –posterior (A-p)

- 1. Anatomical or true conjugate:** from the sacral promontory to the uppermost point of the symphysis pubis. Measures 12cm
- 2. Obstetrical conjugate:** from the **sacral promontory** to the posterior border of the upper surface which is about 1.25cm lower. (The available space for passage of the fetus) 11cm
- 3. Diagonal conjugate:** from the lower border of the symphysis to the sacral promontory.

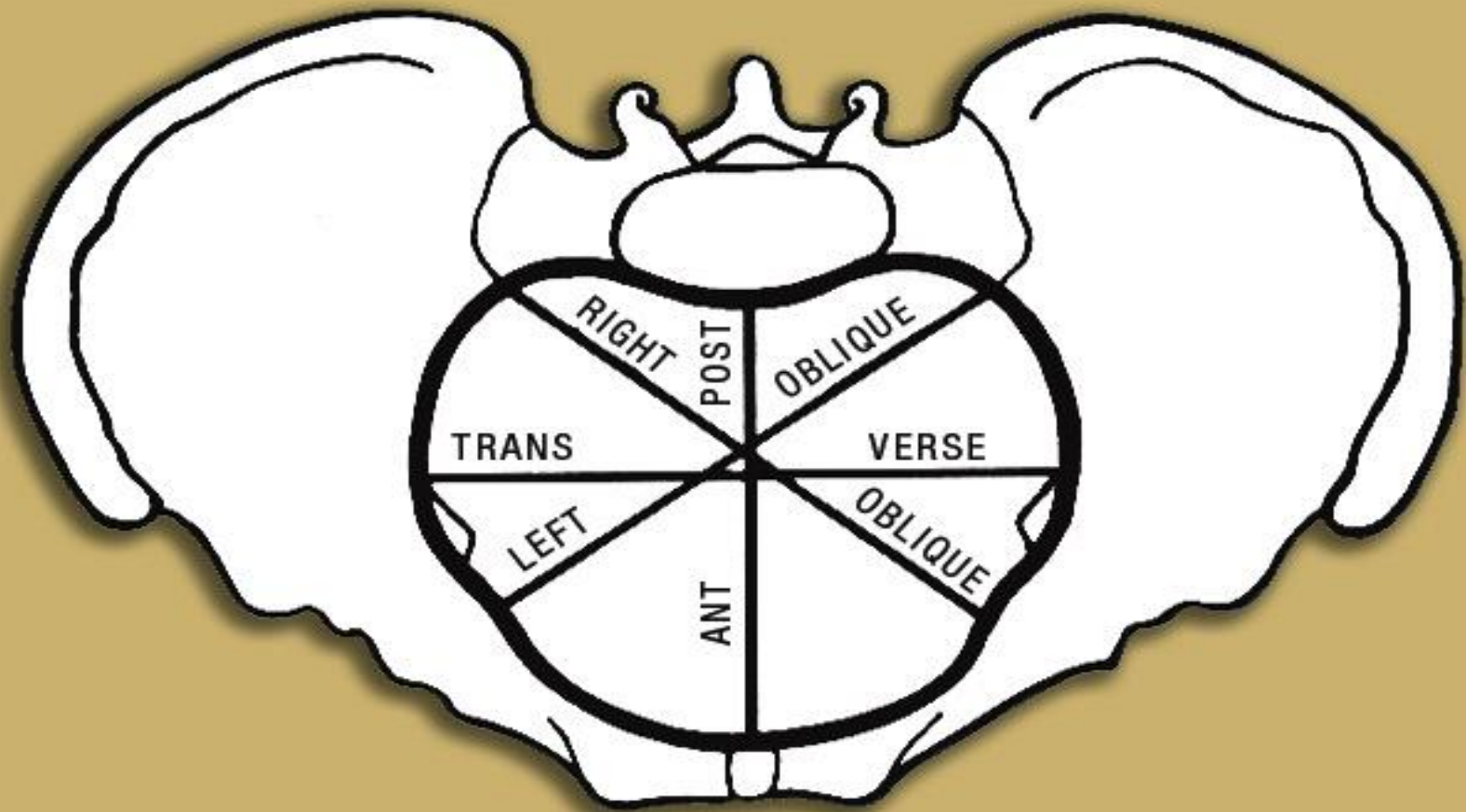
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B. Oblique diameter:-

- Measured from one sacro-iliac joint to ilio-pecteanal eminence of the opposite side # 12cm

C. Transverse diameter:-it is between the farthest points of the iliopectineal line

- The largest diameter of the fetal head enters in this diameter.



Diameters of the pelvic brim

Pelvic cavity

- It is a curved canal
- It extends from the *brim to the outlet*
- circular in shape
- The anterior wall is straight and shallow(Pubic bones and symphysis pubis)
- Its depth is 4cm.
- The posterior wall is deep and concave (the curve of the sacrum) which is 12cm in length.
- Lateral wall: the ischium and parts of the body of the ilium.
- All diametes are 12cm

The pelvic outlet

- Diamond-shape and bounded by:
 - **Anteriorly**: by the arcuate pubic ligaments and the pubic arch.
 - **Laterally**: by ischial tuberosity and the Sacrotuberous ligament
- **Posteriorly** -by the tip of the sacrum

Cont....

- The landmarks of the pelvic outlet are as follows:
 - Lower border of the symphysis pubis
 - Pubic arch
 - Ischial spines and ischial-tuberosities
 - Sacro-tuberous and sacro-spinous ligaments
 - Lower aspect of the sacrum and the coccyx

Anteroposterior

Oblique

Transverse

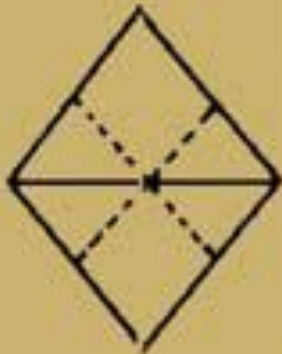
Brim

**11****12****13**

Cavity

**12****12****12**

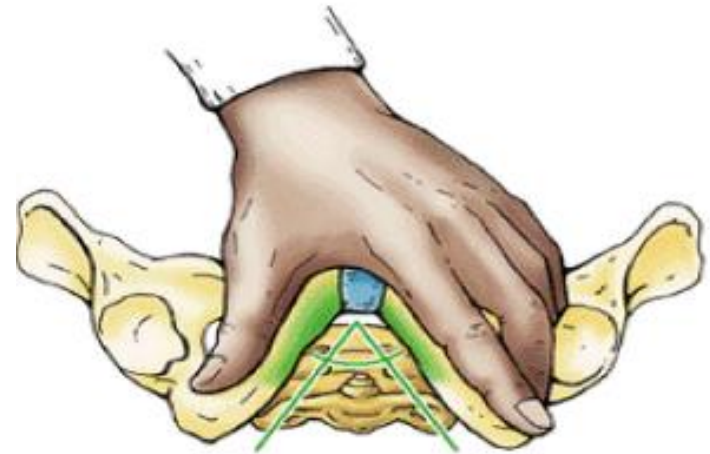
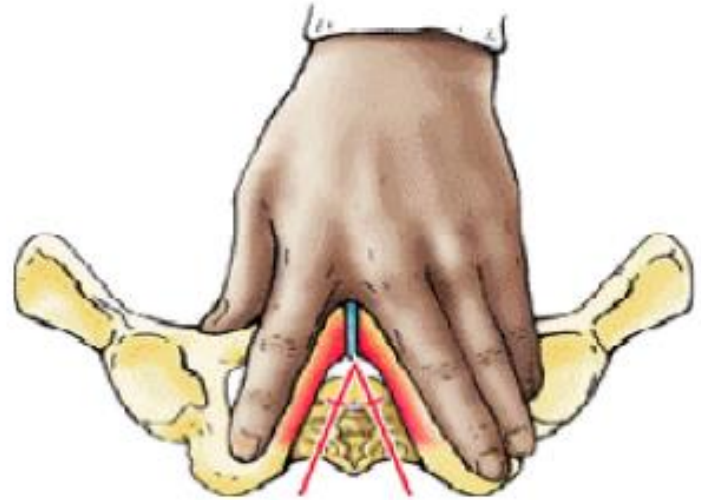
Outlet

**13****12****11**

Pelvic measurements given in centimetres

Sexual differences in bony pelvis

- The male pelvis is heavier and thicker than the female pelvis and usually has more prominent bone markings.
- The female pelvis is wider, shallower, lighter, and rounder than the male and has a larger pelvic inlet and outlet with wider sub-pubic angle



Types of Female Pelvis

They are four types:

1. **Gynecoid pelvis:**-the ideal /classical-pelvis for child bearing.
2. **Android pelvis:** - resemble with the male pelvis.
3. **Anthropoid pelvis:** - Oval brim with A-P is greater than transverse.
4. **Platypelloid pelvis:**-flat with kidney shaped brim.

Shape of bony pelvis

Four broad categories: Gynecoid, anthropoid, android, platypelloid

1. Gynecoid pelvis:

- Classic female shape
- Oval-shaped inlet,
- Diverging midpelvic sidewalls,
- Far-spaced ischial spines

2. Anthropoid pelvis:

- Exaggerated oval shape to the inlet
- Largest diameter being anteroposterior
- Limited anterior capacity to the pelvis
- More often associated with occiput posterior position

Shape of bony pelvis

Four broad categories: Gynecoid, anthropoid, android, platypelloid

3. Android pelvis:

- Male in pattern
- Heart-shaped inlet
- Prominent sacral promontory
- Prominent ischial spines
- Shallow sacrum
- Converging midpelvic sidewalls
- Increased risk of CPD.

4. Platypelloid pelvis:

- Broad, and flat pelvis
- Exaggerated oval-shaped inlet
- Largest diameter being transverse diameter
- Theoretically predisposing to transverse arrest

Clinical pelvimetry

Assessed ≥ 36 and in labor

- Inlet of true pelvis
 - Transverse diameter: largest (>12.0 cm)
 - *True conjugate (obstetric conjugate)*: sacral promontory to superior aspect of symphysis pubis
 - TC= Diagonal C * - 1.5 (to 2.0) cm
 - Normally ≈ 10 to 11 cm

Diagonal conjugate:

- Sacral promontory to inferior margin of the symphysis pubis

Clinical pelvimetry

- Midpelvis
 - Limiting diameter is the interspinous diameter
 - Normally: > 10 cm
- Pelvic outlet
 - Rare clinical significance
 - Anteroposterior diameter: coccyx to symphysis pubis ≈ 13 cm
 - Transverse diameter: inter-tuberosity (ischial) ≈ 8 cm

Clinical pelvimetry

- Favorable pelvic shape for vaginal delivery:
 - Favorable: gynecoid, anthropoid
 - Less favorable: android, platypelloid
 - Many pelvis of women fall into intermediate categories
- Although the assessment of fetal size along with pelvic shape and capacity (x-ray) is still of clinical utility, it is a very inexact science
- An adequate trial of labor is the only definitive method to determine whether a given fetus will be able to safely negotiate a given pelvis

Features	<u>Gynaecoid</u>	Android	Anthropoid	Platypelloid
Brim	Rounded	Heart-shaped	Long oval	Kidney-shaped
Fore pelvis	Generous	Narrow	Narrowed	Wide
Side wall	Straight	Convergent	Divergent	Divergent
Ischial spines	Blunt	Prominent	Blunt	Blunt
Sciatic notch	Rounded	Narrow	Wide	Wide
Sub-pubic angle	90°	$<90^\circ$	$>90^\circ$	$>90^\circ$
Incidence 2/9/2021	50%	20%	25%	5%

Contracted pelvis

- Involves contracture of the inlet, mid cavity & the outlet
- **Inlet contracture:-** A-P diameter of $< 10\text{cm}$ or transverse diameter of $< 12\text{cm}$
- **Mid cavity contracture:-** A-P diameter of < 11.5 or transverse diameter of $< 9\text{cm}$
- **Outlet contracture:-** inter tuberos diameter of $< 8\text{cm}$

Causes of contracted pelvis

- Normal development of pelvic bones but abnormal shape
- Nutritional deficiency from rickets
- Disease or injury on the spine , pelvis , & limbs
- Congenital disorder of spines , pelvis & limbs

Pelvic joints

- There are four pelvic joints: One symphysis pubis, two sacroiliac joints, one sacrococcygeal joint
1. **The symphysis pubis:** is formed at the junction of the two pubic bones which are united by a pad of cartilage.
 2. **The sacroiliac joints:** are the strongest joints in the body. They join the sacrum to the ilium and thus connect the spine to the pelvis
 3. **The sacrococcygeal joint:** is formed where the base of the coccyx articulate with the tip of the sacrum.

The Pelvic Floor:

- It is a muscular diaphragm that separates the pelvic cavity from the perineal space below.

Functions

- ✓ Supports the pelvic viscera
- ✓ To build up effective intra-abdominal pressure
- ✓ It helps the anterior rotation of the presenting part and directs downward and forward along the birth passage
- ✓ Voluntary control of micturation and defecation
- ✓ Plays an important part in sexual intercourse

The Fetal Skull

Introduction

- The fetal skull anatomy and dimensions are very important parameters in labor outcome
- This is because the fetal skull is proportionally the largest and least pliable structure of the fetal anatomy
- Labor obstruction usually occurs at the level of the fetal skull
- Clear understanding of fetal skull anatomy and identification of important landmarks is important in assessment of labor progress and diagnosis of abnormalities of labor

Cont....

- The fetal head is the most difficult part of the fetus whether it comes first or last.
- So understanding of the fetal landmarks & measurement of the fetal skull enables the health professional to recognize normal presentation, position & to facilitate the delivery with the least trauma to mother & child
- The skull is divided into three
 - The vault
 - The bases
 - The face

The Fetal Skull cont...

- **The valut** - is the large domed shaped part above the imaginary line drawn b/n the orbital ridges & the nape of the neck.
- **The base** - is comprised of bones which are firmly united to protect the vital centers in the medulla.
- **The face** - is composed of 14 small bones which are also firmly united & non compressible.

Bones of the valut

There are five bones in the valut of the skull:

- **The occipital bone** - lies at the back of the head & forms the region of the occiput.
- **The two parital bones** - lie on either side of the skull
- **The two frontal bones** - forms the forehead or sinciput

The Fetal Skull cont...

Sutures & fontanelle

- **Sutures** - are cranial joints and are formed where two bones adjoin.
- **Fontanels** - are formed when two or more sutures are meet together

Types of suture

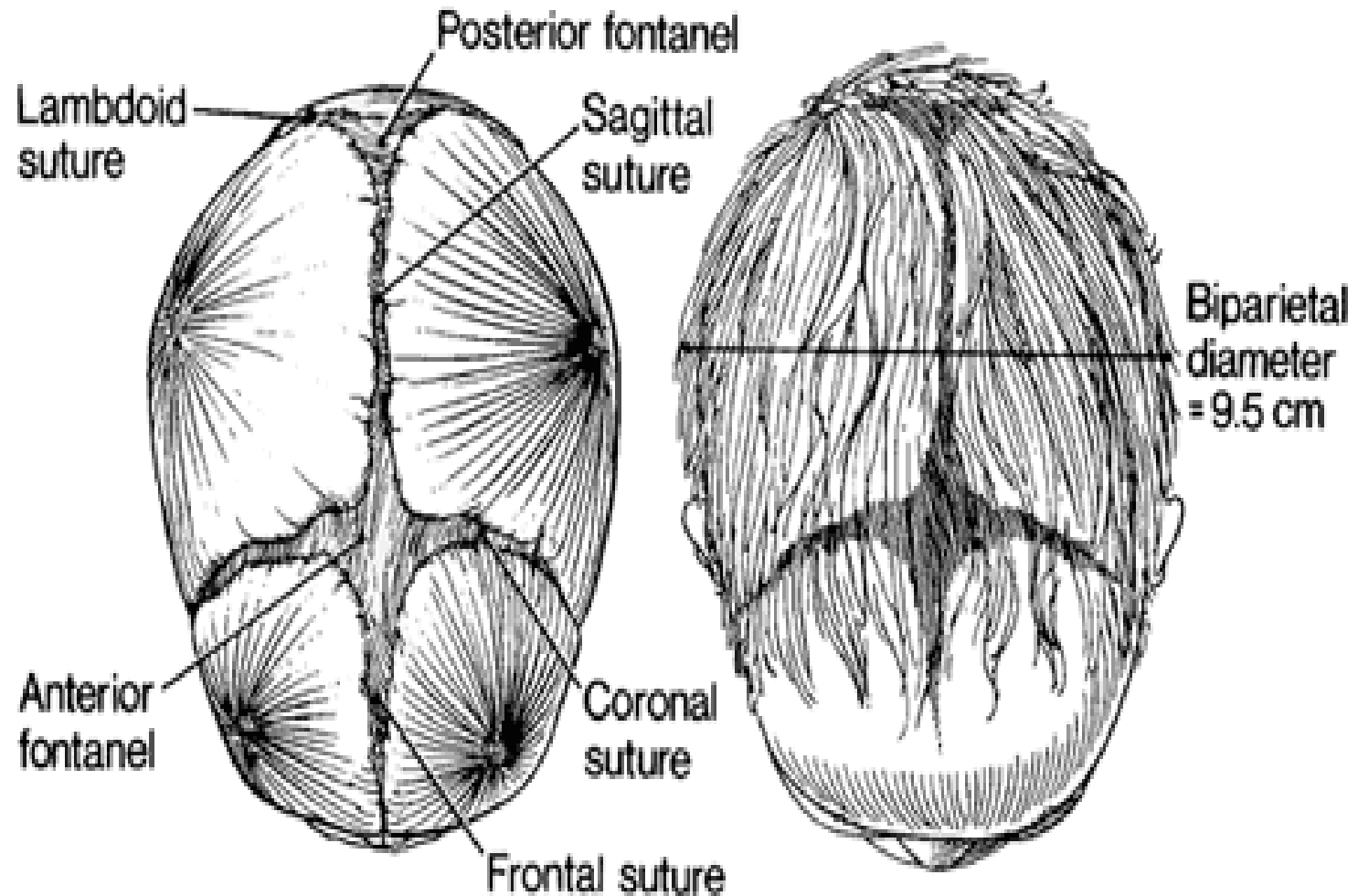
- **The lambdoidal suture** - is shaped like the greek letter lambda & separates the occipital bone from the two parietal bone
- **The sagittal suture** - lie b/n the parietal bones
- **The coronal suture** - separates the frontal bones from the parietal bones.
- **The frontal suture** - runs b/n the two halves of the frontal bone.

The Fetal Skull cont...

Types of fontanels

- **The posterior fontanelle (lambda)** - is situated at the junction of the lambdoidal & sagittal sutures.
- It is small triangular in shape & can be recognized vaginally
- Closes by 6 weeks of age after delivery.
- **The Anterior fontanelle (bregma)** - is found at the junction of the sagittal, coronal & frontal sutures.
- Recognized vaginally, diamond shaped
- Closes at the age of 18 months after delivery

The Fetal Skull cont...



The Fetal Skull cont...

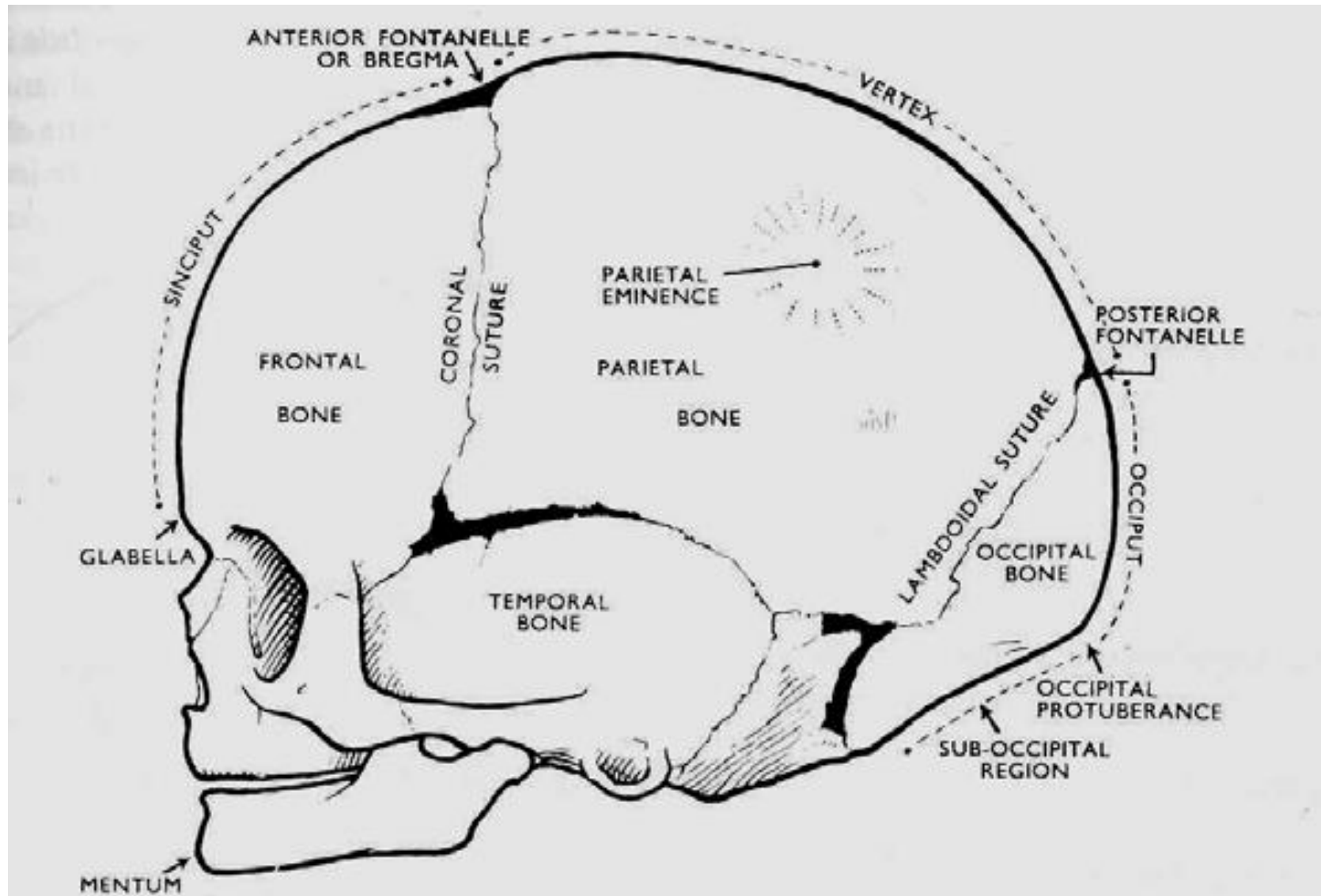
- **Regions of the fetal skull**
- **The occiput** - lies b/n the foramen magnum & the posterior fontanel
- **The vertex** - bounded by the posterior fontanelle, the parital eminence & the anterior fontanelle.
- **The sinciput (brow)** - extends from the anterior fontanelle & the coronal suture to the orbital ridges.
- **The face** - It extends from the orbital ridges & the neck.
- The pint b/n the eye brow is called glabella
- The chin also termed as mentum.

The Fetal Skull cont...

Landmarks of the fetal skull

- The mentum.
- Sinciput
- Anterior fontanelle
- The vertex
- Posterior fontanelle
- Occiput
- Occipital protuberance

The Fetal Skull cont...



Important fetal skull diameters

Diameter	Measurement landmarks	Term fetus	Fetal presentation
Biparietal diameter	Between the two parietal eminences	9.5cm s	Largest transverse diameter of the fetal skull in all cephalic presentations
Sub-occipito bregmatic	Lowest point of occipital bone to the middle of anterior fontanel	9.5cm s	Vertex presentation
Sub-mento bregmatic	Just behind the mentum to the center of anterior fontanel	9 cms	Face presentation
Mento-vertical	Mentum to the highest point of the vertex	13.5 cms	Brow presentation

The BPD is the transverse diameter in all the cephalic presentations including vertex, face and brow as it is the largest transverse diameter of the fetal skull. Measurements represent the averages form a term fetus.

Peculiarities of the Fetal Skull- Landmarks

Fontanels

- **Anterior fontanel (Bregma)** - rhomboid shape space at the junction of the two parietal and frontal bones . Not readily felt during vaginal exam in a well flexed vertex. But easily felt in a brow and face presentation. Differentiated from the posterior fontanel by its four angles
- **Posterior fontanel** - a triangular space at the junction of the occipital and two parietal bones. Has three angles. Useful as the anatomical landmark as the denominator of fetal position in vertex presentations. Easily felt in vertex presentations.

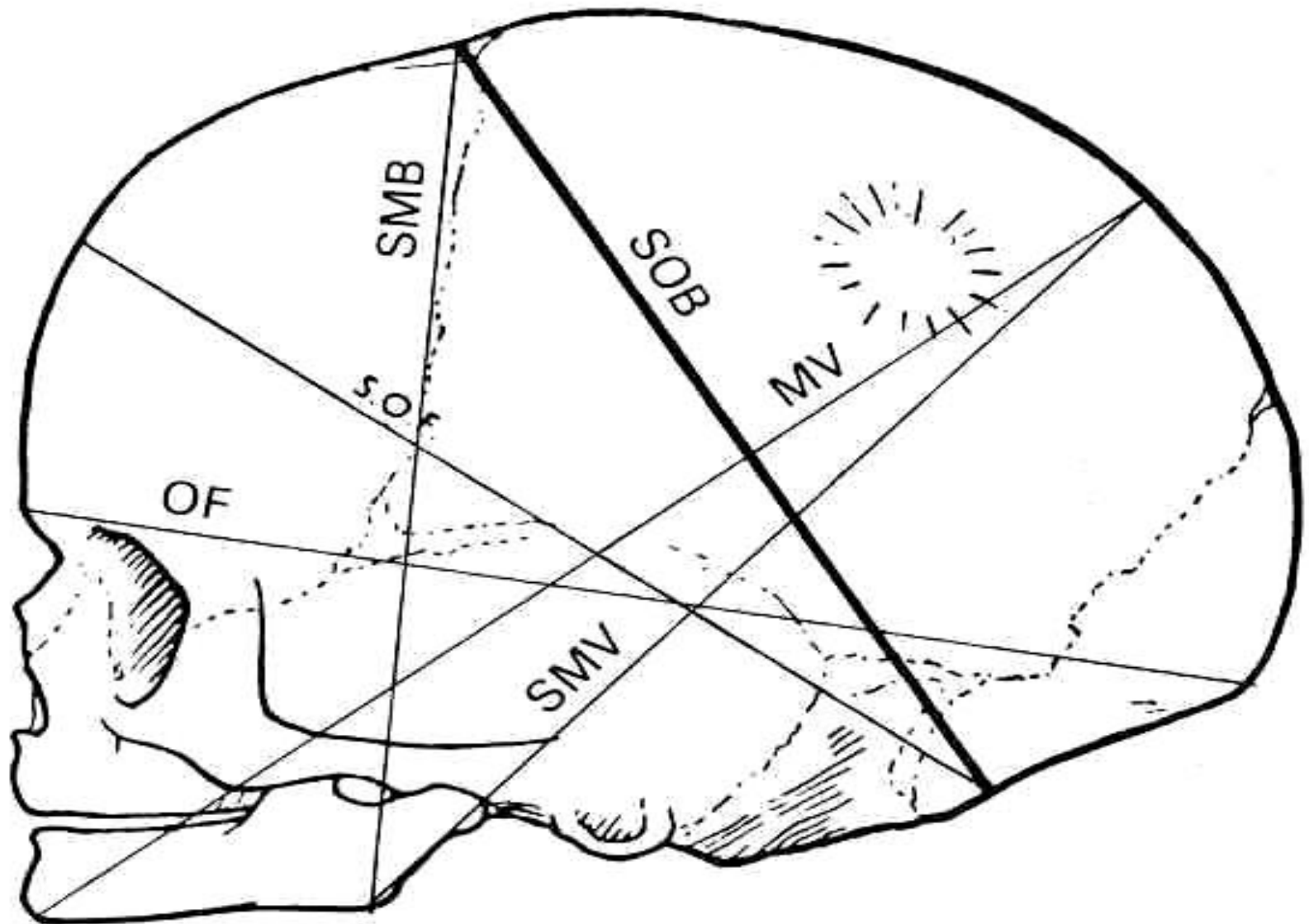
Sutures

- **Sagittal suture** - opening between the two parietal bones. Its central positioning is used to diagnose fetal asynclitism. Also used as the landmark for locating the fontanelles.
- **Lambdoid sutures** - opening between the two parietal bones and the occipital bone posteriorly.
- **Coronal sutures** - opening between the parietal bones and the frontal bone anteriorly.
- Other non-significant sutures as well.
- Significance of these openings is believed to be the provision of pliability of the fetal skull allowing for growth of the fetal and childhood brain.

Transverse diameter

- ***Biparital diameter*** - 9.5cm b/n the parietal eminence
- ***Bitemporal diameter*** - 8.2cm b/n the farthest points of the coronal suture at the temples.

The Fetal Skull cont...



Fetal Molding and Types

- Fetal molding refers to the process of overlap of fetal skull bones on each other at the location of certain sutures usually due to the pressure against the birth canal during labor
- Molding allows for a reduction in fetal skull diameters by upto 6mms to 1 cms which can be significant enough to allow progress to delivery in cases of cephalo-pelvic disproportion
- Excessive molding can also lead to trauma
- Assessment of molding is one of the parameters used to diagnose cephalo-pelvic disproportion
- Two types of molding- parieto-parietal (PP) and occipito-parietal (OP)

Degrees of fetal skull molding during labor

Degree	Description	Significance
0	Skull bones separate from each other	No abnormality
+ 1	Skull bones approximate each other but do not overlap	No abnormality
+ 2	Skull bones overlap but can be separated by the examining hand	Can indicate obstruction if in early labor or at high station
+ 3	Skull bones overlap and cannot be separated by the examining finger	Indicator of obstruction

Caput Succedaneum

- Caput succedaneum is the development of edema on the fetal skull due to the pressure of the cervix impeding venous outflow from the scalp
- It usually recedes hours after delivery
- In cases of obstructed labor caput can grow to excessive sizes and become one of the indicators of obstruction